

Radiometry and Detection Signal Flow in VibeVolts

Simulation Reference Documentation

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1 Introduction

This document outlines the step-by-step mathematical transformations used in the VibeVolts codebase to calculate the target signal, background noise, and resulting Signal-to-Noise Ratio (SNR) on a satellite or observatory detector.

2 Step 1: Solar Flux Calculation

The source of illumination is the Sun. The function `fluxes(band)` in `radiometry_calcs.py` retrieves the solar apparent magnitude m_{sun} and the zero-point calibration flux ZP (flux of a 0-magnitude object in photons/s/m²) for a given band.

The solar flux at Earth/target F_{sun} is calculated as:

$$F_{\text{sun}} = 10^{-0.4 \cdot m_{\text{sun}}} \cdot \text{ZP} \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{m}^2} \right] \quad (1)$$

3 Step 2: Target Reflection (Lambertian Sphere)

The target is modeled as a diffuse Lambertian sphere of radius R and albedo a . The phase angle α is the physical angle between the Sun (light source) and the observer (satellite detector) as seen from the target's center.

The phase function $\Phi(\alpha)$ for a Lambertian sphere is:

$$\Phi(\alpha) = \frac{2}{3\pi^2} [\sin \alpha + (\pi - \alpha) \cos \alpha] \quad (2)$$

The intensity $I(\alpha)$ of the scattered light in the direction of the observer is:

$$I(\alpha) = F_{\text{sun}} \cdot a \cdot (\pi R^2) \cdot \Phi(\alpha) \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{sr}} \right] \quad (3)$$

Inside `lambertian.py`'s `lambertiansphere` function, the returned value (often called `target_brightness` or $J(\alpha)$) is:

$$J(\alpha) = F_{\text{sun}} \cdot \underbrace{\left(a \cdot (\pi R^2) \cdot \frac{2}{3\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha] \right)}_{\text{effective_cross_section}} = \pi \cdot I(\alpha) \quad (4)$$

4 Step 3: Apparent Flux at the Detector

As the reflected light propagates over distance d to the observer, the flux spreads over a sphere's surface. The apparent photon flux (irradiance) at the entrance aperture of the detector, F_{det} , is computed in `scandetectors.py` as:

$$F_{\text{det}} = \frac{J(\alpha)}{\pi d^2} = \frac{I(\alpha)}{d^2} = \frac{2aF_{\text{sun}}R^2}{3\pi d^2} [\sin \alpha + (\pi - \alpha) \cos \alpha] \quad \left[\frac{\text{photons}}{\text{s} \cdot \text{m}^2} \right] \quad (5)$$

5 Step 4: Target Signal Collection

The total number of target signal photons S collected by the sensor during the integration time t is calculated in `scandetectors.py` as:

$$S = F_{\text{det}} \cdot t \cdot A \cdot \eta \cdot f \quad [\text{photons}] \quad (6)$$

Where:

- t : Integration (exposure) time in seconds (`integrationTime`).
- A : Detector aperture entrance area in m^2 (`apertureArea`).
- η : Overall system quantum efficiency (`qe`).
- f : Photometric efficiency fraction inside the photometry bucket (`photoEff`).

6 Step 5: Background Noise Collection

The background sky/space brightness F_{bg} is given in photons/s/sr/ m^2 (derived from the filter band's zero point and magnitudes per square arcsecond).

Assuming background-dominated noise (shot noise), the background photons B collected by a pixel of solid angle Ω (in steradians) is:

$$B = F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta \quad [\text{photons}] \quad (7)$$

The background shot noise N is the standard deviation of this Poisson process:

$$N = \sqrt{B} = \sqrt{F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta} \quad [\text{photons}] \quad (8)$$

7 Step 6: Signal-to-Noise Ratio (SNR) and Integration Time

The SNR γ is:

$$\gamma = \frac{S}{N} = \frac{F_{\text{det}} \cdot t \cdot A \cdot \eta \cdot f}{\sqrt{F_{\text{bg}} \cdot t \cdot A \cdot \Omega \cdot \eta}} = F_{\text{det}} \cdot f \cdot \sqrt{\frac{t \cdot A \cdot \eta}{F_{\text{bg}} \cdot \Omega}} \quad (9)$$

Solving this expression for the required integration time t to achieve a target SNR γ yields:

$$t = \gamma^2 \cdot \frac{F_{\text{bg}} \cdot \Omega}{F_{\text{det}}^2 \cdot A \cdot \eta \cdot f^2} \quad [\text{seconds}] \quad (10)$$

This is the corrected equation implemented in `detector.py`'s `requiredIntegrationTime`.

8 Summary of Variable Definitions

Variable	Code Representation	Description
F_{sun}	<code>sun</code>	Incident solar flux (photons/s/m ²)
R	<code>radius</code>	Radius of the sphere target (m)
a	<code>albedo</code>	Reflected fraction of target surface (0.0...1.0)
α	<code>angle_light_observer</code>	Phase angle (radians)
d	<code>hit_norms</code>	Target-to-detector distance (m)
F_{det}	<code>detectorFlux</code>	Incident target flux on detector (photons/s/m ²)
A	<code>apertureArea</code>	Entrance aperture area (m ²)
t	<code>integrationTime</code>	Exposure time (seconds)
η	<code>qe</code>	System quantum efficiency (0.0...1.0)
f	<code>photoEff / photfrac</code>	Photometric bucket fraction (0.0...1.0)
F_{bg}	<code>space / sky</code>	Background sky radiance (photons/s/sr/m ²)
Ω	<code>pixelOmega</code>	Pixel solid angle (steradians)

Table 1: Variable mappings between the mathematical equations and the python code.